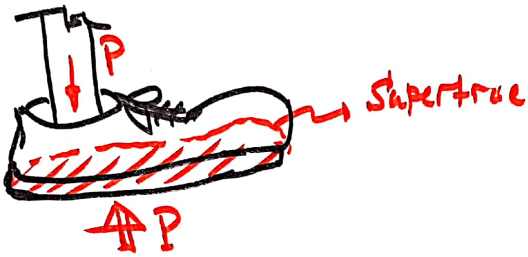
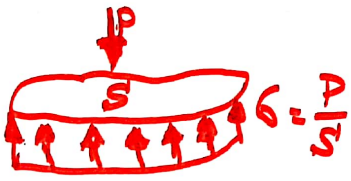
CIMENTACIONES SUPERFICIALES (ZAPATAS)

ZAPATA ⇒ "ZAPATO" (SIMIL)

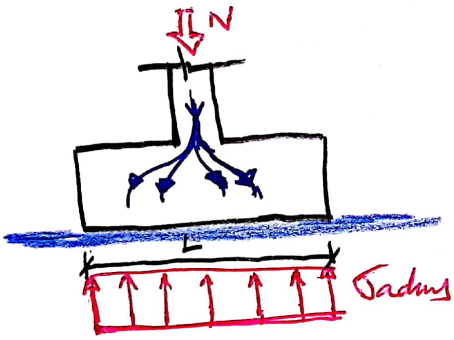


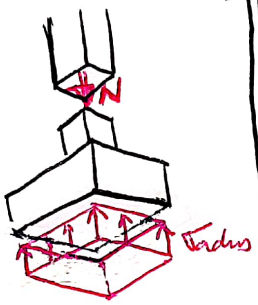


CONCEPTO:

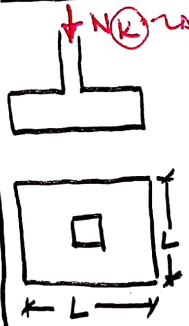
LA FUERZA NORMAL QUE SE TRANSMITE POR EL PILAR NECESITA MAYOR SUPERFICIE PARA QUE CUANDO SE TRANSLADE AL TERRENO NO PRODUZCA TENSIONES MAYORES A LAS ADMISIBLES ($\sigma < \sigma_{ADM}$)

RDAR: EL TERRENO SOPORTA MENOS TENSIÓN QUE EL HORMIGÓN.





EJ: ZAPATA CUADRADA



N_k → (ejemplo) VALOR CARACTERÍSTICO ⇒ (SIN MAYORAR)

- N_k → DATO
- σ_{ADM} → DATO

$$\sigma = \frac{N_k}{S}$$

$$\sigma_{adms} = \frac{N_k}{L^2} \Rightarrow L = \sqrt{\frac{N_k}{\sigma_{adms}}}$$

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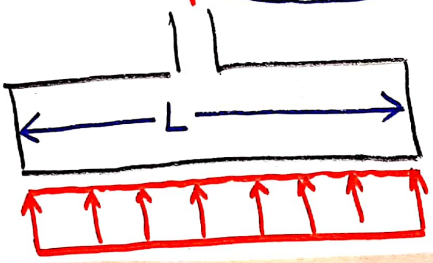
EJEMPLO CONCEPTUAL – CALCULO ÁREA EN PLANTA –

– ZAPATA SOMETIDA ESFUERZO AXIL (PILAR ARTICULADO)

$N_k = 1000\ KN$ (100 tn)



$\sigma_{adms} = \frac{2\ kg}{cm^2} = 20\ tn/m^2 = 0{,}2\ MPa = 200\ Kpa$

¿VALOR L SI ZAPATA ES CUADRADA?

$\sigma = \frac{N}{S} \Rightarrow 20\ \frac{tn}{m^2} = \frac{100\ tn}{L^2} \rightarrow L = \sqrt{\frac{100}{20}} = \sqrt{5} = 2{,}30\ m$

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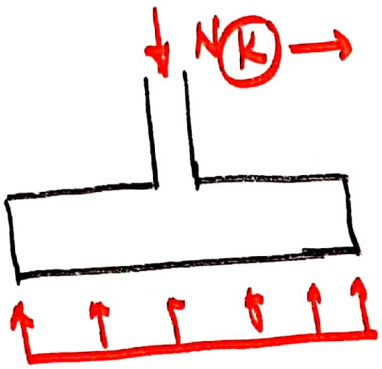
¡OJO! – AHORA AL REVÉS



→ VALOR CARACTERÍSTICO "SIN MAYORAR" ($\gamma_f = 1$)

¡OJO! PARA CALCULAR EL ÁREA DE UNA ZAPATA NO SE MAYORAN LAS ACCIONES Y A ESE LA σ_{ADMS} YA ES EL VALOR MINORADO DE LA RESISTENCIA DEL TERRENO CON UNA MINORACIÓN APROXIMADA DE 3 ($\sigma_{adms} = \frac{\sigma_{HUND}}{3}$)

POR ESTE MOTIVO NO ES NECESARIO MAYORAR ACCIONES CUANDO CALCULAMOS ÁREAS Y TRABAJAMOS CON σ_{ADMS}

$$\frac{N_K}{A} \leq \sigma_{adms} \rightarrow \frac{\sigma_H}{3}$$

$$N_K \leq \sigma_{adms} \cdot L^2$$

$$\boxed{L \geq \sqrt{\frac{\sigma_{adms}}{N_K}}}$$

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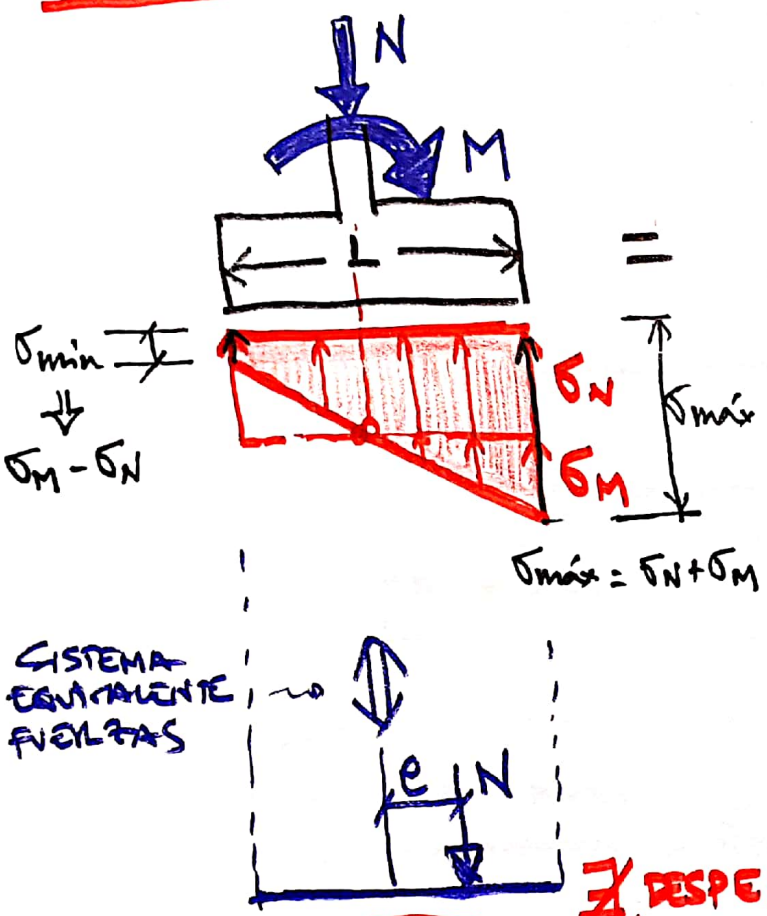
ZAPATA - SOMETIDA (N, M) - ESTRATEGIA ELÁSTICA (ZAPATA CUADRADA)



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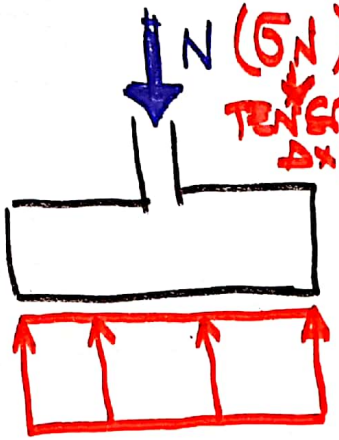
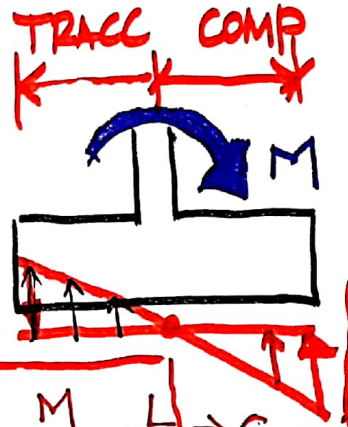


TENSIÓN POR AXIL

$\sigma_N = \frac{N}{A}$

TENSIÓN POR AXIL

+



σ_M

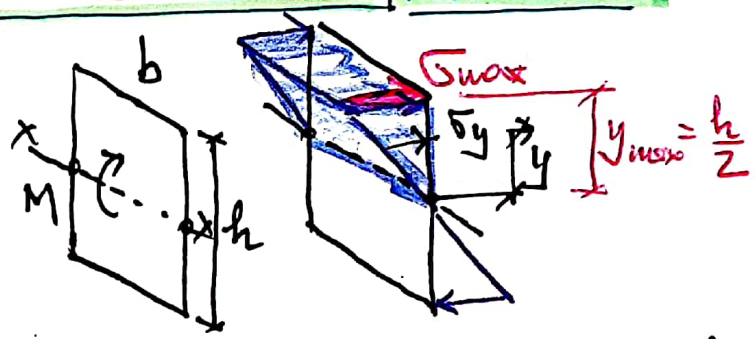
TENSIÓN POR MOMENTO

$$\sigma_M = \frac{M}{\frac{1}{12}L^4} \cdot \frac{L}{2} \Rightarrow \sigma_M = \frac{6M}{L^3}$$

Si $\boxed{e = \frac{M}{N} < \frac{L}{6}}$ ⇒ ∄ DESPEGUE. ∄ TRACCIONES. TODA LA ZAPATA APOYA EN TERRENO

excentricidades pequeñas.

RECORDATORIO



$$\sigma_{máx} = \frac{M}{I}\left(\frac{h}{2}\right) \Leftarrow \begin{cases} \sigma_y = \frac{M_f}{W_{xx}} = \frac{M_f}{I/y} = \frac{M_f \cdot y}{I} \\ I = \frac{1}{12} b h^3 \end{cases}$$

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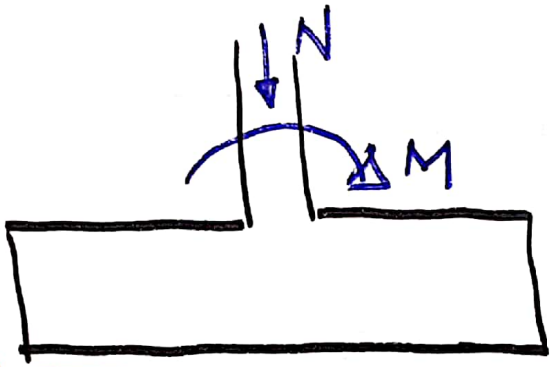
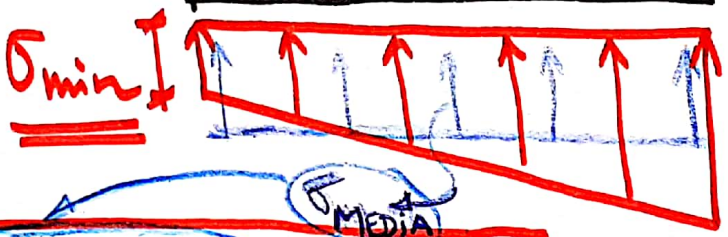
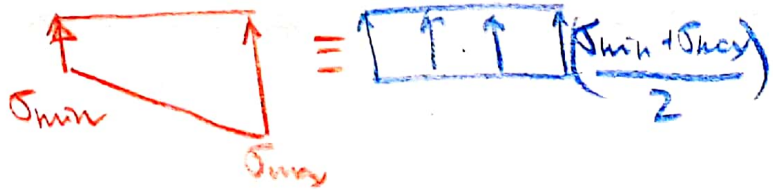
COMPROBACIONES (DATO: INCÓGNITA) ⇒ σ_{adm} terreno





$$\boxed{\sigma_{máx} \leq 1{,}25\ \sigma_{adm}}$$

$$\boxed{\frac{\sigma_{min} + \sigma_{máx}}{2} \leq \sigma_{adm}}$$



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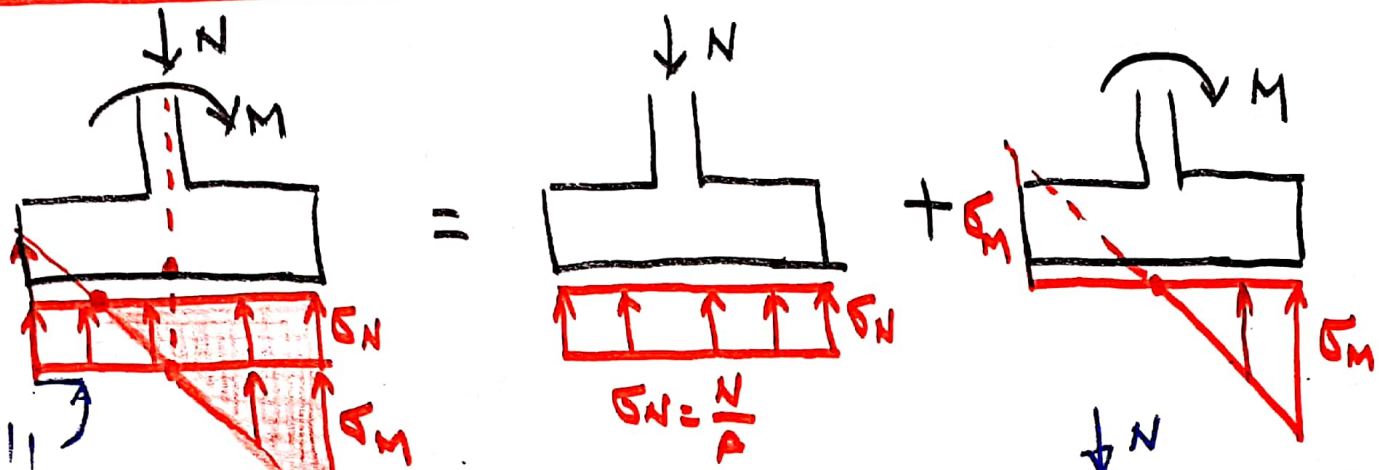
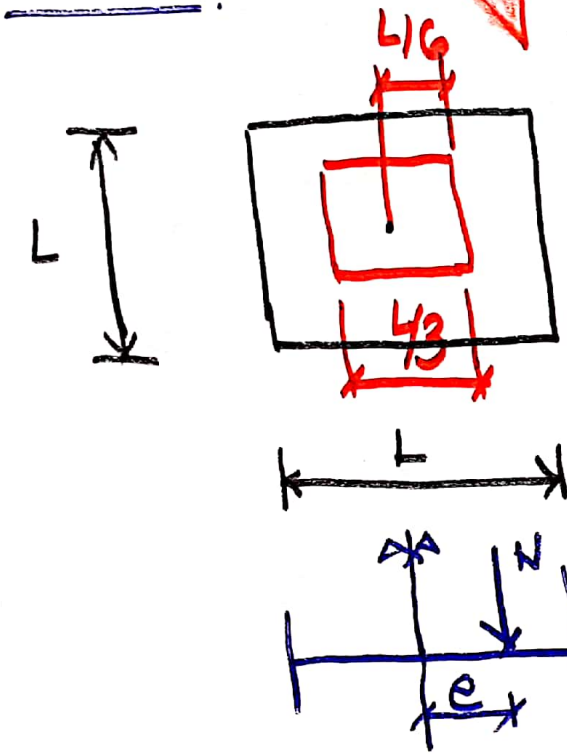
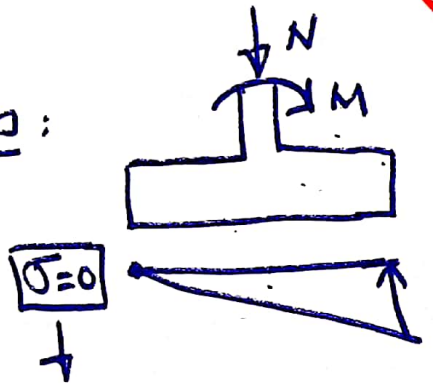
ZAPATA CON MOMENTOS ELEVADOS



DESPEGUE !!



Supongamos:



$\sigma_N = \sigma_M$

$$\frac{N}{L^2} = \frac{6M}{L^3} \Rightarrow \frac{M}{N} = \frac{(1)L^{\cancel{3}}}{6L^{\cancel{2}}} = \frac{L}{6} \Rightarrow \frac{M}{N} = \frac{L}{6}$$

$e \Rightarrow$ excentricidad

$$\boxed{e \leq \frac{L}{6}}$$

"TODO COMPRIMIDO"

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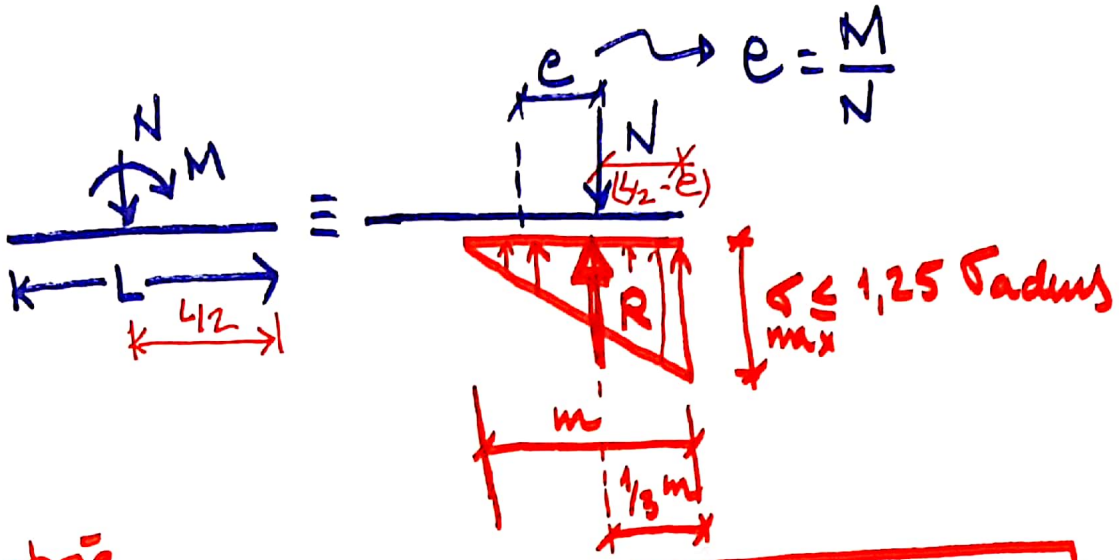
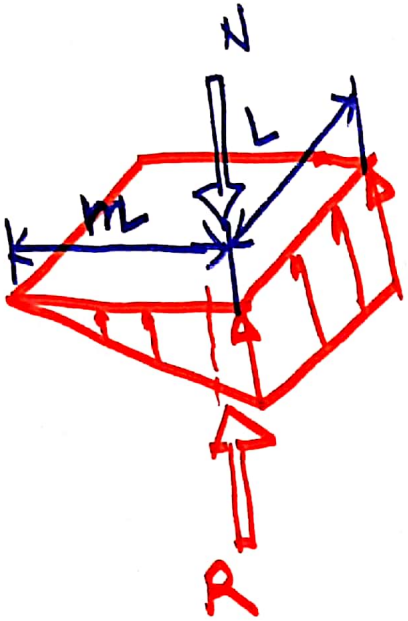
$$\boxed{e \geq \frac{L}{6}} \Rightarrow \text{"DESPEGUE"}$$



$$e = \frac{M}{N}$$

Por geometría al igualar Reacciones N, R $\Rightarrow \frac{L}{2} - e = \frac{m}{3} \rightarrow \boxed{m = 3\left(\frac{L}{2} - e\right)}$



igualamos el volumen de tensiones (R) a la normal (N)

$$\boxed{\frac{1}{2}\, m\, \underbrace{(1{,}25\,\sigma_{adm})}_{\sigma_{max}}\, L = N}$$

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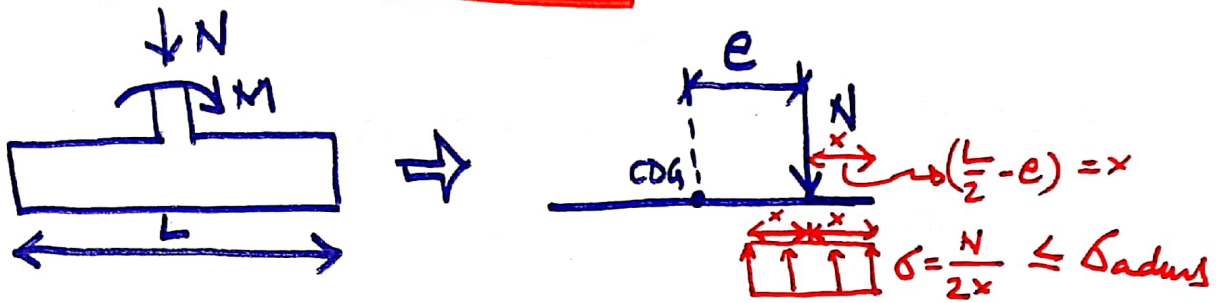
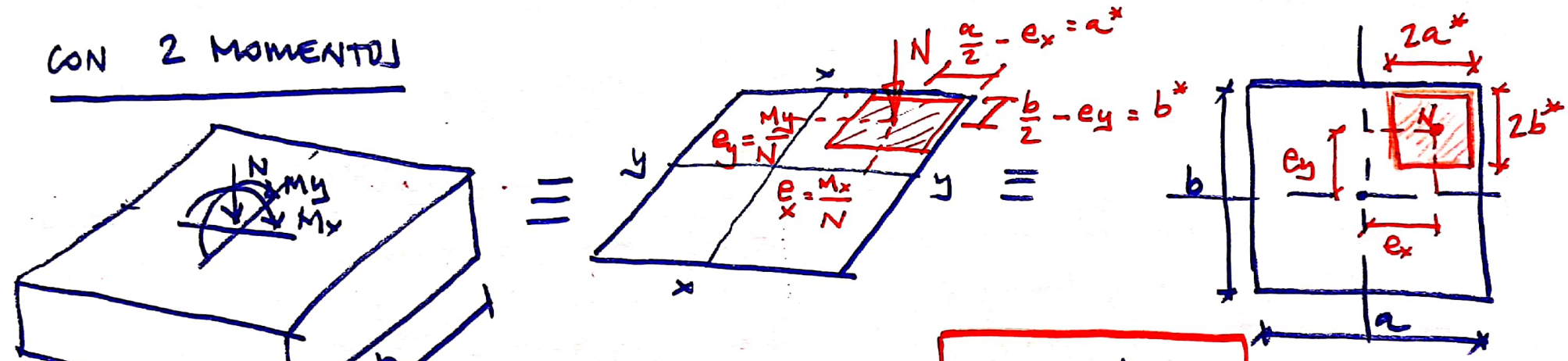
ZAPATAS - SOMETIDAS A 2 MOMENTOS y AXIL (M_x, M_y, N)

(MÉTODO PLÁSTICO) (*) SIMPLE!!

CON 1 MOMENTO



CON 2 MOMENTOS



$$\boxed{\sigma=\frac{N}{(2a^* \times 2b^*)}}$$

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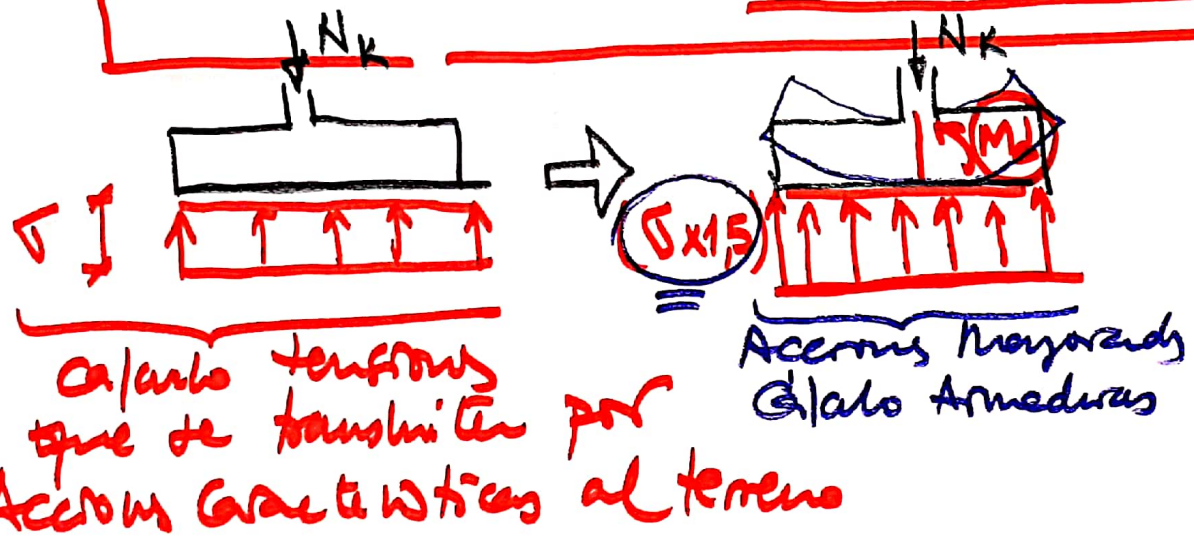
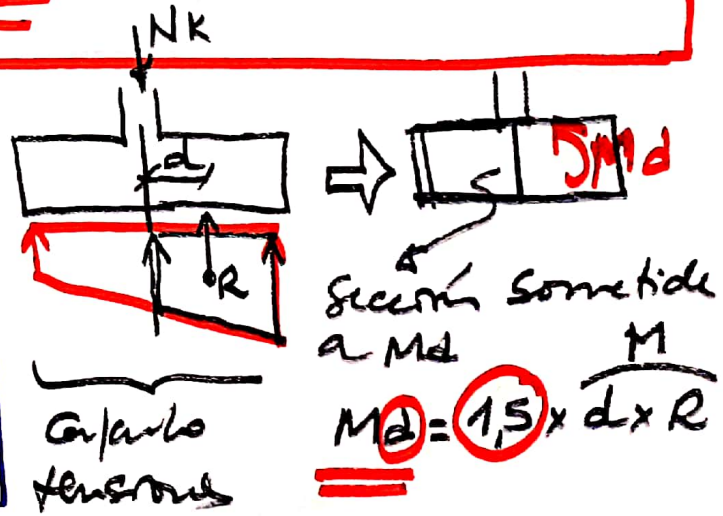
IMPORTANTE ⇒ { TENSIONES / ACCIONES } ⇔ COEF. MAYORACIÓN

NOTA:

PARA EL CÁLCULO DE ARMADURAS (CÁLCULO DE LA ESTRUCTURA DEL CIMIENTO) LAS ACCIONES (TENSIONES DEL TERRENO) SE TIENEN QUE MAYORAR MULTIPLICANDO LAS MISMAS POR COEFICIENTE MAYORACIÓN DE ACCIONES $\gamma_{ACCIONES} \geq 1{,}5$





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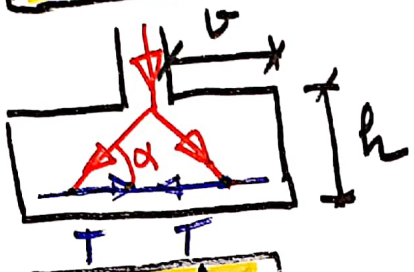
CÁLCULO ARMADURAS EN ZAPATAS

(*) PUNTO DE VISTA ESTRUCTURAL!
(¡ojo! ≠ Rig / Flex Pto vista geotecnico)

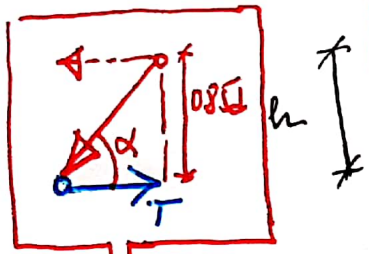
- NO NECESITA ARMADURA CORTANTE
- + VOL. HORMIG
- MENOS DEFORMABLE

(v = vuelo)

RÍGIDA



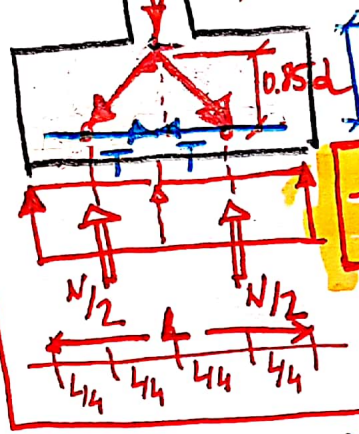
$v < 2h$



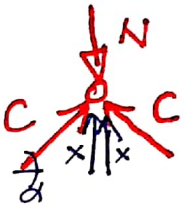
$tg\alpha = 0.85d/(L/4)$

$tg\alpha = 0.85d/0.25L$ (*) 1

$N \cong 1{,}5 N_k = N_d$ (γacciones)



$$T_d = \frac{N_d}{8}\frac{L}{0.85d}$$



$N = 2x = 2C\,sen\alpha$

$N = 2C\,sen\alpha$

$C = \frac{N}{2\,sen\alpha}$ (*) 2

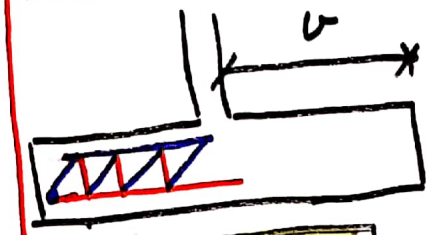
$\cos\alpha = \frac{T}{C} \rightarrow T = C\cdot\cos\alpha$ (*) 2

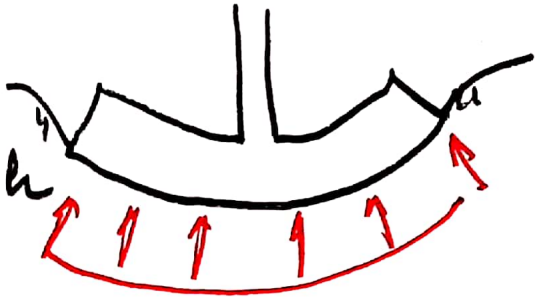
$T = \frac{N}{2\,sen\alpha}\cdot\cos\alpha = \frac{N}{2\,tg\alpha}$

$$T = \frac{N}{2\,tg\alpha} = \frac{N}{2\left(\frac{0.85d}{0.25d}\right)} = \frac{N\cdot L}{8\cdot 0.85d}$$

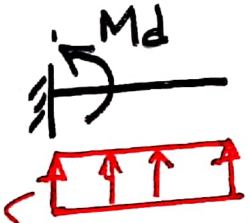
*1

FLEXIBLE





$v > 2h$



$$A_s = \frac{M_d}{0{,}85\,d\cdot f_{yd}}$$

[γadm × γacciones] ≃ 1,5

al calcular las armaduras de la estructura x γacciones (mayoro acciones)